

SECTION 7: INTERNATIONAL TRADE

7.1 GAINS FROM FREE TRADE

MARKING SCHEME

1990/CE/II/42 A	1996/CE/II/53 B	2001/CE/II/44 D	2006/CE/II/44 B (65%)	2013/DSE/I/38 A (50%)
1990/CE/II/48 D	1997/CE/II/53 B	2002/CE/II/44 A (44%)	2006/CE/II/46 A (54%)	2013/DSE/I/39 A (64%)
1991/CE/II/53 A	1998/CE/II/43 B	2002/CE/II/45 C (44%)	2007/CE/II/44 C (67%)	2014/DSE/I/41 A (59%)
1992/CE/II/51 A	1998/CE/II/44 A	2002/CE/II/47 C (69%)	2007/CE/II/45 D (51%)	2014/DSE/I/44 D (39%)
1992/CE/II/53 C	1998/CE/II/45 B	2003/CE/II/45 C (31%)	2007/CE/II/46 B (62%)	2015/DSE/I/40 A (66%)
1993/CE/II/52 A	1999/CE/II/46 A	2003/CE/II/46 B (24%)	2008/CE/II/45 C (65%)	2015/DSE/I/41 D (45%)
1994/CE/II/52 A	1999/CE/II/48 D	2004/CE/II/45 B (50%)	2009/CE/II/46 C (43)	2016/DSE/I/41 B (60%)
1994/CE/II/53 C	2000/CE/II/46 D	2004/CE/II/46 D (40%)	2010/CE/II/45 A (61)	2017/DSE/I/43 B (65%)
1995/CE/II/51 B	2000/CE/II/47 A	2004/CE/II/47 C (65%)	2010/CE/II/46 A (64)	2018/DSE/I/43 D (38%)
1995/CE/II/52 D	2000/CE/II/48 B	2005/CE/II/45 B (47%)	2010/CE/II/47 B (51)	1992/AL/II/17 A
1996/CE/II/46 D	2001/CE/II/42 A	2005/CE/II/46 A (47%)	2012/DSE/I/37 D (65%)	1996/AL/II/25 A
1996/CE/II/47 D	2001/CE/II/43 A	2006/CE/II/43 C (43%)	2012/DSE/I/38 B (47%)	1997/AL/II/15 D
2019/DSE/I/42 C	2020/DSE/I/42 C	2021/DSE/I/43 C	2022/DSE/I/43 B	2023/DSE/I/42 D
2024/DSE/II/42 B	2024/DSE/II/44 D	2025/DSE/I/15 B	2025/DSE/I/42 D	

1991/CE/I/3/(c)

(i) The opportunity cost of producing 1 unit of food

in Country A: $12 \div 6 = 2$ units of clothing

(2)

in Country B: $20 \div 5 = 4$ units of clothing

(2)

Country A has a comparative advantage over Country B in food production because

A produces food at a relatively lower opportunity cost than B.

(3)

(ii) Increase in food = $(6 \text{ units} \times 20) - [(6 \text{ units} \times 10) + (5 \text{ units} \times 10)]$

(1)

= 10 units

(1)

Increase in clothing = $(20 \text{ units} \times 20) - [(12 \text{ units} \times 10) + (20 \text{ units} \times 10)]$

(1)

= 80 units

(1)

(iii) Terms of trade / exchange ratio of trading the two commodities

(2)

1992/CE/1/2/(a)

- (ii) The opportunity cost of producing 1 unit of rice
in Country A is 0.5 units of clothing whereas that (2)
in Country B is 0.75 units of clothing (2)

Country A produces rice at a lower opportunity cost than Country B (1)
 $\therefore$ Country A should specialize in rice production (1)

- (iii) The opportunity cost of Country A in producing 18 ($= 2 \times 9$) of rice is 9 ($= 1 \times 9$) units of clothing (2)
 $\Rightarrow$ Country A gains 1 unit of clothing from every 18 units of rice sold to Country B (2)

OR

The opportunity cost of Country A in producing 10 ($= 1 \times 10$) units of clothing is 20 ($= 2 \times 10$) units of rice (2)
 $\Rightarrow$ Country A saves up 2 units of rice when it buys 10 units of clothing from Country B (2)

- (iv) No, because (1)
the outputs of Country A are doubled but the opportunity cost of producing rice remains the same as in (a)(ii) above (3)
(i.e. 1 unit of rice is exchanged for 0.5 units of clothing).

1993/CE/1/2/(a)

- (i) The opportunity cost of one unit of clothing is 2 units of toys, (1)
while that for toys is 1/2 units of clothing. (1)

No, we can't, because
to find out the comparative advantage, one has to compare the opportunity cost of toy production in the U.S.A with
that in China (and see which cost is lower). (3)

- (ii) The principle of comparative advantage: if each country specializes in producing the good in which it has a lower
opportunity cost, then the total output will increase. (3)
The opportunity cost of producing clothing (or toys) in China is the same as that in the U.S.A., i.e. 1 unit of clothing is
exchanged for 2 units of toys. (2)
 $\Rightarrow$ there is no gain from specialization and trade (1)

- (iii) The new opportunity cost of producing one unit of clothing in China is 3/2 units of toys which is smaller than that (2
units) in the U.S.A. (2)
 $\Rightarrow$ China should specialize in producing clothing and the U.S.A. should specialize in producing toys. Trade between
the two countries can benefit both. (3)

1994/CE/1/9/(c)

Comparative advantage

Unit cost of car production in Japan in terms of apples $= 5/2 (= 2.5)$ units (1)

Unit cost of car production in the U.S.A. in terms of apples $= 4/1 = 4$ units (1)

$\therefore$ Japan has a comparative advantage in producing cars. Conversely, the U.S.A. has a comparative advantage in growing
apples. (1)

OR

Unit cost of apples growing in Japan in terms of cars $= 2/5 (= 0.4)$ units (1)

Unit cost of apples growing in U.S.A. in terms of cars $= 1/4 (= 0.25)$ units (1)

$\therefore$ The U.S.A. has a comparative advantage in growing apples. Conversely, Japan has a comparative advantage in
producing cars. (1)

Gain from trade: Japan

Before specialization (within Japan) 2.5 units of apples (1)

After specialization and trade 3 units of apples (1)

Gain $(3 - 2.5 =) 0.5$ units of apples (1)

Units of apples obtained from 1 unit of cars

Gain from trade: U.S.A.

Before specialization (within the USA) 0.25 units of cars (1)

After specialization and trade 0.3333 units of cars (1)

Gain $(0.3333 - 0.25 =) 0.0833$ units of cars (1)

Units of cars obtained from 1 unit of apples

1995/CE/I/8

- (a) The principle of comparative advantage. (1)
 If each country specializes in producing the good in which it has a lower opportunity cost, then the total output will increase. (3)

- (b) $1.5Y < 1X < 4Y$

OR

$0.25X < 1Y < 0.66X$ (4)

1996/CE/I/10(a)

- (i) Within A, the production cost of 1 unit of clothing is 3 units of rice. (1)
 Within B, the production cost of 1 unit of clothing is $3/2$ units of rice. (1)
 $\therefore$ B has a lower cost or a comparative advantage over A in the production of clothing. (1)

- (ii) (I) Under no trade, the unit price of clothing in A is 3 units of rice forgone. When A trades with B, the import price of 1 unit of clothing is 2 units of rice exported to B.
 $\therefore$ A can save 1 unit of rice per unit of clothing imported from B.

OR

Similarly, the unit price of rice in A before trade is $1/3$ units of clothing. With trade, the export price of 1 unit of rice is $1/2$ units of clothing imported.

$\therefore$ A can gain $1/6$ units of clothing from exporting 1 unit of rice. (3)

- (II) Including the transport cost, the (total) cost of 1 unit of clothing imported from B is 3 units of rice. This equals to the unit cost of production of clothing in A.

$\therefore$ there is no gain for A to trade with B. (3)

1997/CE/I/10(a)

- (i) Same exchange ratio of food and clothing before trade, which is 1 unit of food (F) for 2 units of clothing (C) / same opportunity cost in both countries: $1F = 2C$ (1)+(1)
 $\Rightarrow$ no such terms of trade (1)

- (ii) (I) In Country A, the cost of 1 unit of food is $(20/5 = 4)$ units of clothing. (1)

$\therefore$ In comparison,

[either by showing the cost of 1 unit of food in Country B is 2 units of clothing or explicitly shown in (a)(i)]
 Country B has a lower opportunity cost in producing food. (1)

$\therefore$ Country B has the comparative advantage in food production. (1)

(II)

	Food (units)	Clothing (units)
after trade	-1	+3
before trade	-1	+2
gain from trade		+1

(2)

$\therefore$ 1 unit of clothing is gained. / 0.5 units of food is saved.

(1)

1998/CE/I/10(a)

(i) The opportunity cost of producing 1 unit of computer in

Hong Kong is 10 units of toy (1)

Mainland China is 2.5 units of toy (1)

Taiwan is 5 units of toy (1)

Taiwan has the comparative advantage in producing computers (because her opportunity cost is the lowest among the three economies). (1)

(ii) HK's opportunity cost ($1C = 10T$, $C = \text{clothing}$, $T = \text{toy}$) is the same as the international price ($1C = 10T$) (2)

$\therefore$ HK will not gain. (1)

Mainland China's opportunity cost of producing toys (cost of producing $1T = 0.04C$) is lower than her export price of toys ($1T = 0.1C$) (2)

OR

Mainland China's opportunity cost of producing computers (cost of producing $1C = 25T$) is higher than her import price of computers ($1C = 10T$)

$\therefore$ Mainland China will gain. (1)

1999/CE/I/9(a)

France's opportunity cost of producing one unit of cars is 2 ($= 4/2$) units of red wine. (1)

Japan's opportunity cost of producing one unit of cars is $1/2$ units of red wine.

OR

France's import price of one unit of cars is 1.5 ($= 3/2$) units of red wine. (1)

$\therefore$ France should import cars from Japan. (1)

France's gain = 2 units of red wine - 1.5 units of red wine = 0.5 units of red wine (3)

2000/CE/I/11(a)

(i) Country A can produce $(1\,200 \div 0.4)$ units of wine or $(900 \div 0.6)$ units of clothing.

$\therefore$ A's opportunity cost of wine production is $(900 \div 0.6) / (1\,200 \div 0.4) = 0.5$ units of clothing. (2)

Country B can produce $(400 \div 0.4)$ units of wine or $(2\,400 \div 0.6)$ units of clothing.

$\therefore$ B's opportunity cost of wine production is $(2\,400 \div 0.6) / (400 \div 0.4) = 4$ units of clothing. (2)

A has a lower opportunity cost / comparative advantage in wine production. (1)

$\Rightarrow$ A exports wine. (1)

(ii) No country will gain when the transportation cost per unit of wine $\geq$ total gain of trade per unit of wine / difference in the two domestic opportunity costs. (2)

$\therefore$ min. transportation cost = $[4 - 0.5] = 3.5$ units of clothing (2)

2001/CE/I/9(a)

(i) The opportunity cost of producing 1 unit of garments

Hong Kong: $2 \div 4 = 0.5$ units of computers (1)

USA: $5 \div 4 = 1.25$ units of computers (1)

Hong Kong has a lower opportunity cost in producing garments / comparative advantage in garment production. (1)

$\therefore$ Hong Kong will export garments. (1)

(ii) The range of T.O.T. with transportation is:

1 unit of computers $<$ 1 unit of garments $<$ 1.25 units of computers (3)

2002/CE/I/9

- (a) No, we cannot because (1)
the amount of resources of both countries are not given. (2)
- (b) Yes, we can because (1)
in Country A, the cost of 1 unit of toys is $5/15 = 1/3$ units of computers (1)
in Country B, the cost of 1 unit of toys is $5/10 = 1/2$ units of computers (1)
∴ Country A has a lower cost in producing toys. (1)

2003/CE/I/10(c)

- (i) The opportunity cost of producing 1 unit of paper (1)
in Country A: $10 \div 10 = 1$ unit of cars (1)
in Country B: $30 \div 15 = 2$ units of cars (1)
Country A has a comparative advantage in producing paper, because (1)
its opportunity cost is lower. (1)
Total gain from trading 1 unit of paper for Countries A and B is $[2 - 1 =] 1$ unit of cars (1)
- (ii) No, because (1)
(given that the opportunity cost of 1 unit of paper in Country A is 1 unit of cars while Country X's selling price of 1 unit of paper is 0.5 units of cars)
the lowest possible price which Country A can offer is higher than the price offered by Country X.
- OR**
Country B's gain from importing paper from Country X is greater than the maximum possible gain from importing paper from Country A. (2)

2004/CE/I/11(a)

- (i) No, because (1)
with the same amount of resources, Economy A produces a smaller amount of output than HK. / Economy A has a lower productivity than HK in producing clothing. (1)
- Comparison of output: 10 units vs 12 units **OR** 20 units vs 24 units (1)
- (ii) The opportunity cost of producing one unit of car in Economy A is 2 units of clothing. (1)
The opportunity cost of producing one unit of car in HK is 4 units of clothing. (1)
i.e. lower opportunity cost in Economy A. (1)
- ∴ Economy A has a comparative advantage in producing cars. (1)
- To trade at 5 units of clothing to 2 units of cars, Economy A gains $(5/2 - 2 =) 0.5$ units of clothing from each car exported. (2)

2005/CE/I/10

- (a) (i) Country B. This is because (1)
(with the same amount of resources,) the quantity of output of Country B is larger than that of Country A. (1)
- (ii) - different education and training
- different degree of technological advancement
- different degree of division of labour
- different working environment
- different labour incentives (1@, max: 3)
- [Mark the **FIRST THREE** points only.]

- (b) (i) Principle of comparative advantage: If each country specializes in the production of the good it has a lower opportunity cost, then the total output of all countries will increase. (2)

The opportunity cost of producing 1 unit of watches

in Country A: $5 \div 10 = 0.5$ units of TV sets. (1)

in Country B: $15 \div 20 = 0.75$ units of TV sets. (1)

(i.e., the opportunity cost of producing watches in Country A is lower than that in Country B.)

Therefore, Country A should produce and export watches. (1)

- (ii) (I) The possible range of mutually beneficial terms of trade is
 0.5 units of TV sets $<$ 1 unit of watches $<$ 0.75 units of TV sets (2)

(II) (A country has to compare the terms of trade with its own opportunity cost of production of the good in order to determine whether it should export or import the good.)

If the export price of Country A (the watch exporting country) is lower than its own opportunity cost of production which is 0.5 units of TV sets, Country A would lose from the trade.

OR

If the import price of Country B (the watch importing country) is higher than its own opportunity cost of production which is 0.75 units of TV sets, Country B would lose from the trade. (4)

2006/CE/1/8

- (a) The opportunity cost in the production of 1 unit of clothing

in Country A is $200 \div 100 = 2$ units of computers (1)

in Country B is $450 \div 150 = 3$ units of computers (1)

Therefore, Country A has the comparative advantage in producing clothing. (1)

- (b) Country A gains $2.5 - 2$ (1)
 $= 0.5$ units of computers from 1 unit of export. (1)

2007/CE/1/10(b)

- (i) $(10 - 5)$ units of vegetables (1)
 $= 5$ units of vegetables (2)

- (ii) 5 units of vegetables $<$ 1 unit of jewels $<$ 9 units of vegetables
OR
 $1/9$ units of jewels $<$ 1 unit of vegetables $<$ $1/5$ units of jewels (3)

2008/CE/1/7

- (a) No because (1)
they have the same amount of output with the same amount of resources. (2)

- (b) The opportunity cost of producing food

in Country A $= 50 \div 20 = 2.5$ units of machines (1)

in Country B $= 60 \div 40 = 1.5$ units of machines (1)

Yes because (1)
the opportunity cost of producing food in Country B is lower. (1)

2009/CE/1/10(b)

- (i) The opportunity cost of producing one unit of computers

in Mainland China: $90 \div 50 = 1.8$ units of toys (1)

in Taiwan: $250 \div 200 = 1.25$ units of toys (1)

Taiwan enjoys a comparative advantage in producing computers, because she has a lower opportunity cost in producing computers. (1)

- (ii) Gain of Taiwan from exporting a unit of computers $= (1.5 - 1.25 - 0.2)$ (1)
 $= 0.05$ units of toys (2)

2010/CE/I/10(b)

- (i) If each economy specializes to produce the good in which it enjoys a comparative advantage in production / lower opportunity cost of production, the total output of all economies will increase. (2)
- (ii) In HK: the opportunity cost of producing 1 unit of medicine is 2.5 units of watches.
In Japan: the opportunity cost of producing 1 unit of medicine is 2 units of watches.

The possible range of mutually beneficial terms of trade is
2 units of watches < 1 unit of medicine < 2.5 units of watches (4)

2012/DSE/II/9

- (a) In country A, opportunity cost of producing 1S = 0.2W (1)
In country B, opportunity cost of producing 1S = 0.25W (1)
- (b) If each country specializes in producing the good in which the country has a lower opportunity cost, the total output will increase. (2)
Country A would export shirts. (1)
- (c) $0.2W < 1S < 0.25W$ OR $4S < 1W < 5S$ (2)

2013/DSE/II/5

- (a) In Country A, the opportunity cost of producing 1 smartphone (S) = 0.75 Food (F)
In Country B, the opportunity cost of producing 1 smartphone (S) = 0.5 Food (F)
Country B has a comparative advantage in producing smartphones. (3)
- (b) Yes, because the potential gain from trade $(0.75F - 0.5F = 0.25F)$ per unit of smartphones is greater than the transportation cost (0.2F). (3)

2014/DSE/II/8

- (a) In Country A, the opportunity cost in producing 1 unit of watches (W) = 4 units of Paper (P).
In Country B, the opportunity cost in producing 1 unit of watches (W) = 2 units of Paper (P).
Country B will specialize in producing watches and export watches because the opportunity cost of producing watches is lower there than in Country A. (3)
- (b) The terms of trade is $1W = 2.5P$. Per unit of watches imported, Country A will gain 1.5 P (= $4P - 2.5P$). (3)

2015/DSE/II/7

- (a) Uncertain. Since the amount of resources owned by the two countries is not given, their productivities (i.e., output-input ratios) cannot be determined and compared. (2)
- (b) In Country A, the opportunity cost in producing 1 unit of clothing (C) in terms of rice (R) = $15/3 = 5$ units of rice (R).
In Country B, the opportunity cost in producing 1 unit of clothing (C) in terms of rice (R) = $8/4 = 2$ units of rice (R).
Therefore Country A (B) enjoys comparative advantage in producing rice (clothing). (2)

Per unit gain from trade for Country B = $4R - 2R = 2R$
Total gain from trade = $(4R - 2R) \times 3 = 6R$

OR

In other words, Country B would be the clothing-exporting country, producing 4C, exporting 3C in exchange for 12R and keeping 1C for domestic consumption.
Without trade, it could have produced 1 C and 6R on its own. Its total gain from trade is thus 6R (= $12R - 6R$). (2)

2016/DSE/II/7

- (a) In Country A, the opportunity cost of producing 1C = $6/2 = 3T$
 The terms of trade: 1C will exchange for $(11/2 =) 5.5T$
 As the opportunity cost of producing 1 C is lower than the terms of trade, Country A will produce and export C. (3)
- (b) The gain from trade = $5.5 - 3 = 2 = 0.5T$ per unit of C (3)

2017/DSE/II/7

- (a) (i) Uncertain. Without knowing the total amount of resources of both countries, the productivity (output per unit of resource) for food and computer cannot be determined. (2)
- (ii) Total producible outputs in the two countries when they utilize all of their resources in producing either food or computer:

	Food (units)	Computer (units)
Country A	200	100
Country B	200	200

- The opportunity cost of producing 1 unit of computer in Country A:
 $= 200/100 = 2$ units of food (1)
- The opportunity cost of producing 1 unit of computer in Country B:
 $= 200/200 = 1$ unit of food (1)
- Country B has a lower opportunity cost in producing computers, so it will produce and export computers. (1)
- (b) Terms of Trade (TOT) = $7/4 = 1.75 F$ per unit of C (1)
 Per unit gain for Country B = $(1.75 - 1)F = 0.75F$ (1)
 Total gain = $0.75 \times 16 = 12F$ (1)

2018/DSE/II/8

- (a) The opportunity cost of producing 1T in China = 5C
 The opportunity cost of producing 1T in Japan = 2C (2)
- (b) If each country specialises in producing the good in which the country has lower opportunity cost, then total world output would increase. Japan would export toys, as it has a lower cost of producing toys than China. (3)
- (c) Range of TOT: $2.05C < 1T < 4.95C$ (2)

2019/DSE/II/8

- (a) (i) Yes. (2)
 When Country B has the same resource as Country A, the output of cars would be 200 units which is higher than that of country A.

- (ii) Yes. (3)
 The opportunity cost of producing 1C in Country A = 1T
 The opportunity cost of producing 1C in Country B = 0.6T
 Country B has a comparative advantage in producing cars.

7A) Country B. The production cost of toy is lower than country A. (0.5 car lower than 1 car)

B. Terms of trade 5 toys for 4 cars = 1 toy for 0.8 car. (1 mark)

Exporting country gain = Terms of trade - exporting country production cost - transportation cost.
 = 0.8 car - 0.5 car - 0.1 car = 0.2 car. (2 marks)

In Country A, the opportunity cost of producing 1W = $10/4 = 2.5$ R

In Country B, the opportunity cost of producing 1W = $6/3 = 2$ R

Range of terms of trade: $2R < 1W < 2.5R$

OR

In Country A, the opportunity cost of producing 1R = $4/10 = 0.4$ W

In Country B, the opportunity cost of producing 1R = $3/6 = 0.5$ W

Range of terms of trade: $0.4W < 1R < 0.5W$

- (a) The opportunity cost of producing 1R in Country E=2T (4)

The opportunity cost of producing 1R in Country F=3T

Since Country E has a lower opportunity cost in producing rice, Country E exports rice and Country F imports rice.

Gains from trade for Country E=2.2T-2T=0.2T

Gains from trade for Country F=3T-2.2T=0.8T

- (b) Here, Country F has no absolute advantage in producing either good, and yet it has (2)

Comparative advantage in producing toys. As shown in (a), Country F can still enjoy gains from trade(=0.8T per 1R) by exporting toys in exchange for rice from Country E.

- (a) (i) No, because with the same amount of resources, Country A can only produce the same number of computers as Country B. (2)

(ii) Yes.

Opportunity cost of producing 1 unit of computers:

in Country A = $500/800 = 0.625$ units of rice (1)

in Country B = $300/400 = 0.75$ units of rice (1)

As Country A can produce computers at a lower opportunity cost than Country (1)

B, Country A has a comparative advantage in producing computers.

- (b) 0.625 units of rice < 1 unit of computers < 0.7 units of rice (2)

- (a) Absolute advantage cannot be determined (1)
 as the actual amounts of resources in the two countries are not known. (1)

- (b) Outputs when each country allocates all its resources to producing either bread or clothing:

	Bread (units)		Clothing (units)
Country X	$100 / 0.25 = 400$	OR	$75 / 0.75 = 100$
Country Y	$80 / 0.25 = 320$	OR	$240 / 0.75 = 320$

Hence, opportunity cost to produce 1 unit of bread (2)
 in Country X = $100/400 = 0.25$ units of clothing
 in Country Y = $320/320 = 1$ unit of clothing

As Country X has a lower cost in producing bread, it is the bread-exporting country.

The total gain from trade from exporting 30 units of bread (1)
 $= 30 \times (7/10 - 0.25)$ (1)
 $= 13.5$ units of clothing